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1. Which of the following are legal entry point methods that can be run from the command line? (Choose all that apply.)
- A. `private static void main(String[] args)`
 - B. `public static final main(String[] args)`
 - C. `public void main(String[] args)`
 - D. `public static final void main(String[] args)`
 - E. `public static void main(String[] args)`
 - F. `public static main(String[] args)`

Busque las 4 palabras clave: public, static, void, main y elegí las opciones que las cumplieran

2. Which answer options represent the order in which the following statements can be assembled into a program that will compile successfully? (Choose all that apply.)
- X: `class Rabbit {}`
Y: `import java.util.*;`
Z: `package animals;`
- A. X, Y, Z
 - B. Y, Z, X
 - C. Z, Y, X
 - D. Y, X
 - E. Z, X
 - F. X, Z
 - G. None of the above

Siguiendo el orden: package, import y class, las demás aunque no tengan package o import las tome en cuenta ya que siguen un orden correcto

3. Which of the following are true? (Choose all that apply.)

```
public class Bunny {  
    public static void main(String[] x) {  
        Bunny bun = new Bunny();  
    }  
}
```

- A. **Bunny is a class.**
- B. bun is a class.
- C. main is a class.
- D. Bunny is a reference to an object.
- E. **bun is a reference to an object.**
- F. main is a reference to an object.
- G. The main() method doesn't run because the parameter name is incorrect.

Siguiendo lo que vimos las primeras clases, Bunny es el nombre de la clase, mientras que bun es un objeto al que se está apuntando en memoria

4. Which of the following are valid Java identifiers? (Choose all that apply.)

- A. _
- B. **_helloWorld\$**
- C. true
- D. java.lang
- E. **Public**
- F. 1980_s
- G. **_Q2_**

Con las reglas de “_” no se puede usar como identificador, pero si para inicializar el nombre de uno, Public no es reservada pero public si

bro

5. Which statements about the following program are correct? (Choose all that apply.)

```
2: public class Bear {  
3:     private Bear pandaBear;  
4:     private void roar(Bear b) {  
5:         System.out.println("Roar!");  
6:         pandaBear = b;  
7:     }  
8:     public static void main(String[] args) {  
9:         Bear brownBear = new Bear();  
10:        Bear polarBear = new Bear();  
11:        brownBear.roar(polarBear);  
12:        polarBear = null;  
13:        brownBear = null;  
14:        System.gc(); } }
```

- A. The object created on line 9 is eligible for garbage collection after line 13.
- B. The object created on line 9 is eligible for garbage collection after line 14.
- C. The object created on line 10 is eligible for garbage collection after line 12.
- D. The object created on line 10 is eligible for garbage collection after line 13.
- E. Garbage collection is guaranteed to run.
- F. Garbage collection might or might not run.
- G. The code does not compile.

El objeto construido no se destruye hasta que brownBear es nulo, ya que rompe la conexión que existe entre ambos objetos y garbage collection los destruye.

6. Assuming the following class compiles, how many variables defined in the class or method are in scope on the line marked on line 14?

```
1: public class Camel {  
2:     { int hairs = 3_000_0; }  
3:     long water, air=2;  
4:     boolean twoHumps = true;  
5:     public void spit(float distance) {  
6:         var path = "";
```

```
7:      { double teeth = 32 + distance++; }
8:      while(water > 0) {
9:          int age = twoHumps ? 1 : 2;
10:         short i=-1;
11:         for(i=0; i<10; i++) {
12:             var Private = 2;
13:         }
14:         // SCOPE
15:     }
16: }
17: }
```

- A. 2
- B. 3
- C. 4
- D. 5
- E. 6
- F. 7**
- G. None of the above

La línea 2 no se toma en cuenta ya que es un bloque sin acceso, lo mismo para la línea 7, por ultimo las variables del for al estar únicamente existiendo dentro de el, no tiene alcance fuera de los corchetes.

7. Which are true about this code? (Choose all that apply.)

```
public class KitchenSink {  
    private int numForks;  
  
    public static void main(String[] args) {  
        int numKnives;  
        System.out.print("""  
            "# forks = " + numForks +  
            " # knives = " + numKnives +  
            "# cups = 0""");  
    }  
}
```

- A. The output includes: # forks = 0.
- B. The output includes: # knives = 0.
- C. The output includes: # cups = 0.
- D. The output includes a blank line.
- E. The output includes one or more lines that begin with whitespace.
- F. The code does not compile.

No se evalúan las variables y Java toma todo como un string por las tripes comillas

8. Which of the following code snippets about var compile without issue when used in a method? (Choose all that apply.)

- A. var spring = null;
- B. var fall = "leaves";
- C. var evening = 2; evening = null;
- D. var night = Integer.valueOf(3);
- E. var day = 1/0;
- F. var winter = 12, cold;
- G. var fall = 2, autumn = 2;
- H. var morning = ""; morning = null;

Var, admite string, valores numéricos, por eso la opción E es valida, tengo entendido que Java toma únicamente el valor asignado, por eso la opción D es válida

9. Which of the following are correct? (Choose all that apply.)
- A. An instance variable of type `float` defaults to `0`.
 - B. An instance variable of type `char` defaults to `null`.
 - C. A local variable of type `double` defaults to `0.0`.
 - D. A local variable of type `int` defaults to `null`.
 - E. A class variable of type `String` defaults to `null`.
 - F. A class variable of type `String` defaults to the empty string `""`.
 - G. None of the above.

Las variables locales no tienen valores predeterminados, una variable `float` debe tener punto decimal, por lo que la opción E es correcta

10. Which of the following expressions, when inserted independently into the blank line, allow the code to compile? (Choose all that apply.)

```
public void printMagicData() {  
    var magic = _____;  
    System.out.println(magic);  
}
```

- A. `3_1`
- B. `1_329_.0`
- C. `3_13.0_`
- D. `5_291._2`
- E. `2_234.0_0`
- F. `9___6`
- G. `_1_3_5_0`

Un “`_`” puede usarse como separador entre números, pero no se puede iniciar una variable con `_`, ni usarse antes o después de un punto

11. Given the following two class files, what is the maximum number of imports that can be removed and have the code still compile?

```
// Water.java  
package aquarium;  
public class Water { }
```

```
// Tank.java
package aquarium;
import java.lang.*;
import java.lang.System;
import aquarium.Water;
import aquarium.*;
public class Tank {
    public void print(Water water) {
        System.out.println(water); } }
```

- A. 0
- B. 1
- C. 2
- D. 3**
- E. 4
- F. Does not compile

Ambas java.lang es redundante importarlas si ya se hace por defecto, lo mismo para Aquarium, no es necesario conservar ambos, por ese motivo solo es necesario quedarse con un import

12. Which statements about the following class are correct? (Choose all that apply.)

```
1: public class ClownFish {
2:     int gills = 0, double weight=2;
3:     { int fins = gills; }
4:     void print(int length = 3) {
5:         System.out.println(gills);
6:         System.out.println(weight);
7:         System.out.println(fins);
8:         System.out.println(length);
9:     } }
```

- A. Line 2 generates a compiler error.**
- B. Line 3 generates a compiler error.
- C. Line 4 generates a compiler error.**
- D. Line 7 generates a compiler error.**
- E. The code prints 0.
- F. The code prints 2.0.
- G. The code prints 2.
- H. The code prints 3.

No se puede inicializar dos variables solo con una coma de separación, tampoco dentro de un método y fins está bloqueada al estar entre los corchetes generando un bloque.

13. Given the following classes, which of the following snippets can independently be inserted in place of INSERT IMPORTS HERE and have the code compile? (Choose all that apply.)

```
package aquarium;  
public class Water {  
    boolean salty = false;  
}
```

```
package aquarium.jellies;  
public class Water {  
    boolean salty = true;  
}
```

```
package employee;  
INSERT IMPORTS HERE  
public class WaterFiller {  
    Water water;  
}
```

- A. `import aquarium.*;`
- B. `import aquarium.Water;`
`import aquarium.jellies.*;`
- C. `import aquarium.*;`
`import aquarium.jellies.Water;`
- D. `import aquarium.*;`
`import aquarium.jellies.*;`
- E. `import aquarium.Water;`
`import aquarium.jellies.Water;`
- F. None of these imports can make the code compile.

No se puede usar dos * en una importación y al extraer dos veces Water es redundante y java marca un error

14. Which of the following statements about the code snippet are true? (Choose all that apply.)

```
3: short numPets = 5L;  
4: int numGrains = 2.0;  
5: String name = "Scruffy";  
6: int d = numPets.length();  
7: int e = numGrains.length;  
8: int f = name.length();
```


- A. Line 3 generates a compiler error.
- B. Line 4 generates a compiler error.
- C. Line 5 generates a compiler error.
- D. Line 6 generates a compiler error.
- E. Line 7 generates a compiler error.
- F. Line 8 generates a compiler error.

Java se enfoca en el tipo de dato y no en el valor de la variable, no se puede declarar un int con un punto decimal y en la línea 7 existe un error de sintaxis

15. Which of the following statements about garbage collection are correct? (Choose all that apply.)
- A. Calling `System.gc()` is guaranteed to free up memory by destroying objects eligible for garbage collection.
 - B. Garbage collection runs on a set schedule.
 - C. Garbage collection allows the JVM to reclaim memory for other objects.
 - D. Garbage collection runs when your program has used up half the available memory.
 - E. An object may be eligible for garbage collection but never removed from the heap.
 - F. An object is eligible for garbage collection once no references to it are accessible in the program.
 - G. Marking a variable `final` means its associated object will never be garbage collected.

Uno de sus principales motivos es liberar espacio, además de eliminar los objetos en memoria cuando ya no hay apuntadores o se pierde la conexión por completo

16. Which are true about this code? (Choose all that apply.)

```
var blocky = ""
    squirrel \s
    pigeon  \
    termite""";
System.out.print(blocky);
```

- A. It outputs two lines.
- B. It outputs three lines.
- C. It outputs four lines.
- D. There is one line with trailing whitespace.
- E. There are two lines with trailing whitespace.
- F. If we indented each line five characters, it would change the output.

El programa inicializa en squirrel y pigeon marcando dos lineas en la salida del programa

17. What lines are printed by the following program? (Choose all that apply.)

```
1: public class WaterBottle {  
2:     private String brand;  
3:     private boolean empty;  
4:     public static float code;  
5:     public static void main(String[] args) {  
6:         WaterBottle wb = new WaterBottle();  
  
7:         System.out.println("Empty = " + wb.empty);  
8:         System.out.println("Brand = " + wb.brand);  
9:         System.out.println("Code = " + code);  
10:    } }
```

- A. Line 8 generates a compiler error.
- B. Line 9 generates a compiler error.
- C. Empty =
- D. Empty = false
- E. Brand =
- F. Brand = null
- G. Code = 0.0
- H. Code = 0f

El valor por defecto en un string es null, en un boolean es false y en un float es 0.0

18. Which of the following statements about var are true? (Choose all that apply.)

- A. A var can be used as a constructor parameter.
- B. The type of a var is known at compile time.
- C. A var cannot be used as an instance variable.
- D. A var can be used in a multiple variable assignment statement.
- E. The value of a var cannot change at runtime.
- F. The type of a var cannot change at runtime.
- G. The word var is a reserved word in Java.

Usando las reglas de var obtuve que no se pueden usar como instancias, no puede ser modificado una vez que se declara, no es una palabra reservada dentro de java, es incorrecto usar var en una instrucción de múltiples variables así que las opciones correctas son B C F

19. Which are true about the following code? (Choose all that apply.)

```
var num1 = Long.parseLong("100");  
var num2 = Long.valueOf("100");  
System.out.println(Long.max(num1, num2));
```

- A. The output is 100.
- B. The output is 200.
- C. The code does not compile.
- D. num1 is a primitive.
- E. num2 is a primitive.

Para num1 parseLong devuelve un primitivo por lo que D es correcto, num 2 al pasar por valueOf se convierte en un no primitivo, pero al compararse Java lo convierte a primitivo para poder hacer la comparación, la respuesta correcta es A

20. Which statements about the following class are correct? (Choose all that apply.)

```
1: public class PoliceBox {  
2:     String color;  
3:     long age;  
4:     public void PoliceBox() {  
5:         color = "blue";  
6:         age = 1200;  
7:     }  
8: }
```

```

7:     }
8:     public static void main(String []time) {
9:         var p = new PoliceBox();
10:        var q = new PoliceBox();
11:        p.color = "green";
12:        p.age = 1400;
13:        p = q;
14:        System.out.println("Q1="+q.color);
15:        System.out.println("Q2="+q.age);
16:        System.out.println("P1="+p.color);
17:        System.out.println("P2="+p.age);
18:    } }

```

- A. It prints Q1=blue.
- B. It prints Q2=1200.
- C. It prints P1=null.**
- D. It prints P2=1400.
- E. Line 4 does not compile.
- F. Line 12 does not compile.
- G. Line 13 does not compile.
- H. None of the above.

Se genero un método en lugar de un constructor public void PoliceBox() por lo que el código no se va a ejecutar con el new, mandando valores por defecto

21. What is the output of executing the following class?

```

1: public class Salmon {
2:     int count;
3:     { System.out.print(count+"-"); }
4:     { count++; }
5:     public Salmon() {
6:         count = 4;
7:         System.out.print(2+"-");
8:     }
9:     public static void main(String[] args) {
10:        System.out.print(7+"-");
11:        var s = new Salmon();
12:        System.out.print(s.count+"-"); } }

```

- A. 7-0-2-1-
- B. 7-0-1-
- C. 0-7-2-1-
- D. 7-0-2-4-**
- E. 0-7-1-
- F. The class does not compile because of line 3.
- G. The class does not compile because of line 4.
- H. None of the above.

Tomando el orden correcto de los bloques, obtenemos que primero se ejecuta los bloques estáticos, bloque de instancia y por ultimo el constructor

22. Given the following class, which of the following lines of code can independently replace INSERT CODE HERE to make the code compile? (Choose all that apply.)

```
public class Price {
    public void admission() {
        INSERT CODE HERE
        System.out.print(amount);
    }
}
```

- A. `int Amount = 0b11;`
- B. `int amount = 9L;`
- C. `int amount = 0xE;`
- D. `int amount = 1_2.0;`
- E. `double amount = 1_0_.0;`
- F. `int amount = 0b101;`**
- G. `double amount = 9_2.1_2;`**
- H. `double amount = 1_2_.0_0;`

Nuevamente es correcto utilizar “_” para separar números, es correcto declarar un variable binaria con un int, “_” no puede ir antes o después de un punto, no se puede declarar un int con un float, no Amont porque es error de sintaxis

23. Which statements about the following class are true? (Choose all that apply.)

```
1: public class River {  
2:     int Depth = 1;  
3:     float temp = 50.0;  
4:     public void flow() {  
5:         for (int i = 0; i < 1; i++) {  
6:             int depth = 2;  
7:             depth++;  
8:             temp--;  
9:         }  
  
10:        System.out.println(depth);  
11:        System.out.println(temp); }  
12:    public static void main(String... s) {  
13:        new River().flow();  
14:    } }
```

- A. Line 3 generates a compiler error.**
- B. Line 6 generates a compiler error.
- C. Line 7 generates a compiler error.
- D. Line 10 generates a compiler error.**
- E. The program prints 3 on line 10.
- F. The program prints 4 on line 10.
- G. The program prints 50.0 on line 11.
- H. The program prints 49.0 on line 11.

Tengo entendido que al declarar un float debe agregarse una f al final por lo que seria error de sintaxis, depth esta declarado dentro de un for por lo que fuera del corchete no se tiene acceso